

EE309 CPU PROJECT

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1 Modifications from multicycle

- We have made five stages in pipeline each stage performs its task and then stores the value to be given to next stage in the register pillars.
- We control enable signal of registers and make it one during falling edge of clk.
- Each of the five stages is used during one clock cycle, unlike multicycle, which uses one stage during one cycle.
- We have used ring buffer for instruction and data memory.

2 Implementation

We implemented the ISA provided, comprising 8 instructions (add,sub,mul,addi,sll,lw,sw,jri). We used the provided testbench and modified it to validate the working of each instruction. This was done whilst keeping in mind of not running into any RAW data hazard.

By storing register values in Data memory and then loading them to other registers, we validated the working of lw/sw instructions and the cyclic shifting of the ring buffer. We have defined a PC(initialized to 0) that keeps track of which instruction we are currently executing. As one can see from the waveform, JRI instruction "jumps" PC by the specified amount whilst all other instructions just increment it by 2.

Below We have provided the IMEM that we initialized via testbench and the final outputs of Data Memory and registers R0-R7.

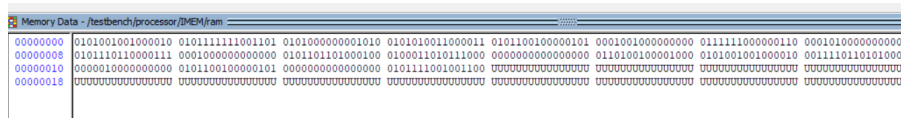


Figure 1: IMEM

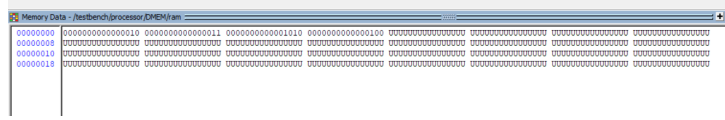


Figure 2: DMEM

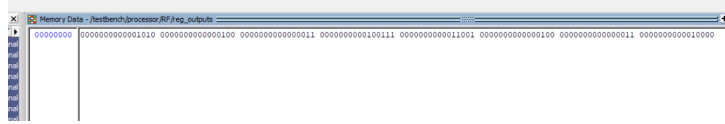


Figure 3: Registers

3 Comparison of Time required

We synchronized the clocks and used the same set of five instructions in data and instruction memory, noting the time difference in execution.

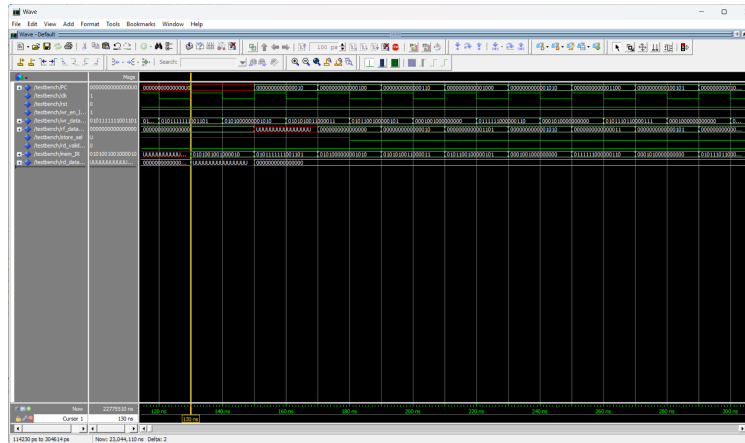


Figure 4: Pipelining start time

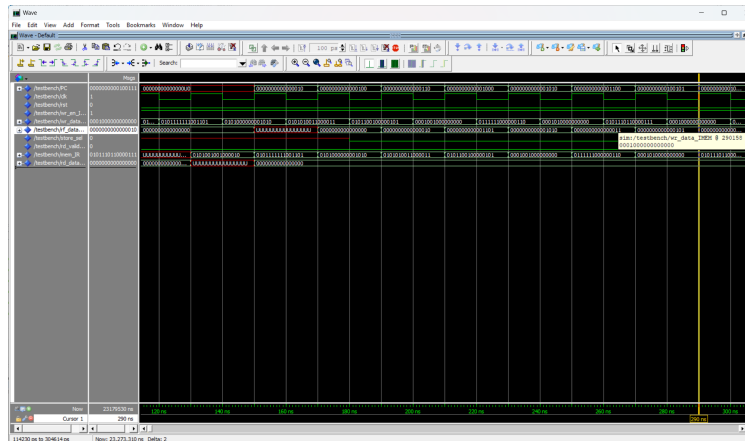


Figure 5: Pipelining end time

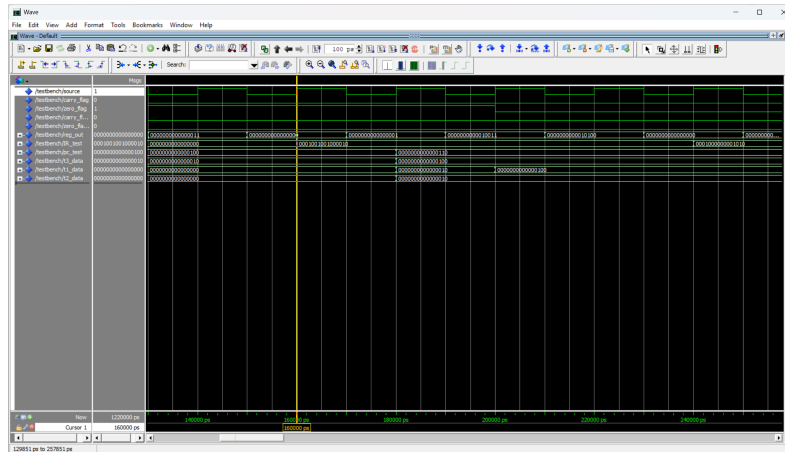


Figure 6: Multicycle start time

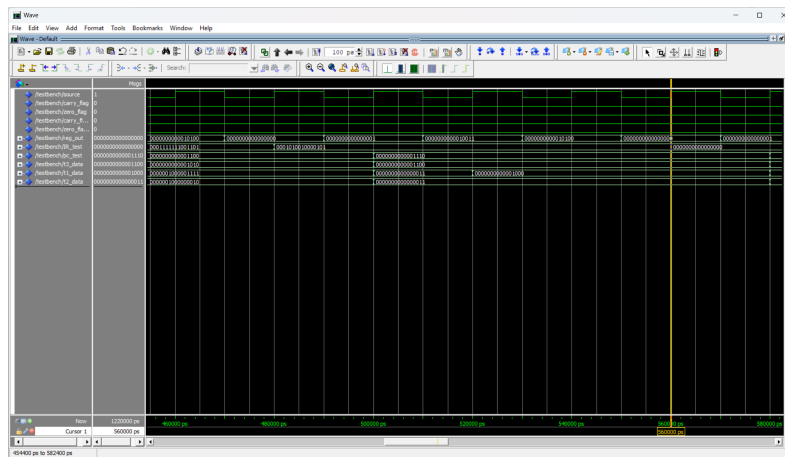


Figure 7: Multicycle end time

Time required by pipelining for executing five instructions= 160 ns

Time required by multicycle for executing five instructions= 400 ns

This shows implementing pipeline has reduced the execution time significantly.